Comprehensive Study Notes: Eigenvalues and Eigenvectors

Welcome to the world of Eigenvalues and Eigenvectors! As a first-year Computer Science engineering student, you will use these concepts everywhere—from computer graphics and machine learning (like Principal Component Analysis) to Google's PageRank algorithm.

These notes are designed for an absolute beginner. We will break down what these terms mean, why they matter, and exactly how to calculate them, step-by-step.

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1. High-Level Overview: What are they?

The Intuition

In linear algebra, a **Matrix (A)** acts like a transformation machine. When you multiply a matrix by a vector (X), the matrix usually changes the vector's direction and length.

However, for any given matrix, there are a few special, "magical" vectors. When the matrix multiplies these special vectors, **their direction does not change**. They only get stretched, shrunk, or flipped along their original line.

- **Eigenvector (X):** The special vector whose direction remains unchanged after being transformed by the matrix. (Note: The zero vector cannot be an eigenvector).
- **Eigenvalue (λ):** The scaling factor. It is the exact amount by which the eigenvector is stretched or shrunk.

The Core Equation

Mathematically, this relationship is written as:

$$AX = \lambda X$$

- A = Your square matrix ($n \times n$)
- X = The Eigenvector (a non-zero column vector)
- λ (Lambda) = The Eigenvalue (a scalar number, like 2, -1, or 5)

Diagram: A 2D coordinate plane showing three vectors. A red vector X1 and a green vector X2 are shown. The transformation matrix A is applied to both. The new vector AX1 points in a completely different, non-parallel direction (purple). However, the new vector AX2 (orange) points in the exact same direction as the original green vector X2, just stretched longer. This visual proves X2 is an eigenvector because it remained parallel to itself, while X1 is not.

Visual Description for TikZ Generation: Create a standard 2D Cartesian coordinate system with X and Y axes. Draw a red vector $\vec{X}_1 = (1,0)^T$ along the x-axis. Draw a green vector $\vec{X}_2 = (0,1)^T$ along the y-axis. Draw a purple vector $A\vec{X}_1 = (1,2)^T$ starting from the origin. Draw an orange vector $A\vec{X}_2 = (0,4)^T$ starting from the origin and lying directly on top of the y-axis, extending past the green vector. Add labels for all four vectors. Include a brief caption stating that $\vec{X}_2$ and $A\vec{X}_2$ are parallel, proving $\vec{X}_2$ is an eigenvector.

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2. The Math: How to Find Them (Step-by-Step)

If we want to find these values, we start with the core equation and use simple algebra.

Step A: Rearrange the Equation

$$AX = \lambda X$$

Move everything to one side:

$$AX - \lambda X = 0$$

To factor out the X, we can't just subtract a number (λ) from a matrix (A). We must multiply λ by the **Identity Matrix (I)** (a matrix with 1s on the diagonal and 0s everywhere else) so it becomes a matrix of the same size as A:

$$(A - \lambda I)X = 0$$

Step B: The Characteristic Equation (Finding the Eigenvalue)
In the equation $(A - \lambda I)X = 0$, we know X is an eigenvector, so it **cannot be zero**.
For this equation to have a non-zero solution for X, the matrix $(A - \lambda I)$ must collapse space, meaning it does not have an inverse. In linear algebra, a matrix has no inverse if its **determinant is exactly zero**.

This gives us the most important formula, called the **Characteristic Equation**:

$$|A - \lambda I| = 0$$

Solving this determinant will give you a polynomial equation. The roots (solutions) of this polynomial are your **Eigenvalues (λ)**.

Step C: Finding the Eigenvectors
Once you have found your Eigenvalues, you plug them back into the equation $(A - \lambda I)X = 0$ one by one, and solve for X using row reduction (Gauss Elimination).

> 🔥 **Pro-Tip: Shortcut for 3x3 Matrices**
> Expanding a 3×3 determinant manually can be tedious. You can directly write the characteristic polynomial using this shortcut:
>
> $$\lambda^3 - \text{Trace}(A)\lambda^2 + (\text{Sum of minors of main diagonal})\lambda - |A| = 0$$
>
> - **Trace(A):** The sum of the elements on the main diagonal of A.
> - **Minor:** The determinant left over when you cross out the row and column of a specific element.
> - **|A|:** The determinant of matrix A.

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3. Key Properties for Quick Problem Solving

Memorize these properties; they are heavily tested in exams to see if you can skip long calculations.

Property	Explanation
Transpose Equality	A matrix A and its transpose A^T have the exactly same eigenvalues.
Triangular Matrices	If a matrix is upper or lower triangular, its eigenvalues are simply the numbers sitting on its main diagonal.
Sum = Trace	The sum of all eigenvalues equals the sum of the main diagonal elements (the Trace). $\lambda_1 + \lambda_2 + ... = a_{11} + a_{22} + \dots$
Product = Determinant	The product of all eigenvalues equals the determinant of the matrix. $\lambda_1 \times \lambda_2 \times ... = \|A\|$
Inverses	If A has an eigenvalue λ, then its inverse matrix A^{-1} has the eigenvalue $\frac{1}{\lambda}$.
Powers	If A has an eigenvalue λ, then the matrix A^m has the eigenvalue λ^m.
Symmetric Matrices	A matrix where $A = A^T$ will *always* have purely real eigenvalues (no imaginary numbers).

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4. The Cayley-Hamilton Theorem

This theorem is a brilliant mathematical trick used to find the inverse of large matrices easily.

The Theorem States:

> *Every square matrix satisfies its own characteristic equation.*

What does this mean?
If your characteristic equation for matrix A is:

$$\lambda^2 - 4\lambda - 5 = 0$$

You can replace every λ with the matrix A itself (and attach the Identity matrix I to the constant):

$$A^2 - 4A - 5I = 0$$

How to use it to find an Inverse (A^{-1}):

1. Write out the theorem equation: $A^2 - 4A - 5I = 0$
2. Multiply the entire equation by A^{-1}:

$$A^{-1}(A^2 - 4A - 5I) = A^{-1}(0)$$

$$A - 4I - 5A^{-1} = 0$$

3. Rearrange to solve for A^{-1}:

$$5A^{-1} = A - 4I \implies A^{-1} = \frac{1}{5}(A - 4I)$$

Now you can find an inverse using simple subtraction rather than complex row operations!

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5. Worked Examples (Step-by-Step)

Let's walk through how to solve these problems exactly as you should on an exam.

Example 1: Finding Eigenvalues & Vectors for a 2x2 Matrix

Problem: Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$.

Step 1: The Characteristic Equation $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

Multiply the diagonals to find the determinant:

$$(1-\lambda)(3-\lambda) - (4)(2) = 0$$

$$3 - \lambda - 3\lambda + \lambda^2 - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

Factoring this quadratic equation gives $(\lambda - 5)(\lambda + 1) = 0$.
Result: The Eigenvalues are $\lambda = 5$ and $\lambda = -1$.

Step 2: Find the Eigenvector for $\lambda = -1$
Set up $(A - \lambda I)X = 0$. Substitute $\lambda = -1$:

$$\begin{bmatrix} 1-(-1) & 2 \\ 4 & 3-(-1) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Notice the second row is just a multiple of the first. If we reduce it, we just get one useful equation:

$$2x + 2y = 0 \implies x + y = 0 \implies y = -x$$

Let x be any constant k. Then $y = -k$.
Eigenvector for $\lambda = -1$ is $X = k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. (Usually, we just write the base vector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$).

Step 3: Find the Eigenvector for $\lambda = 5$
Substitute $\lambda = 5$:

$$\begin{bmatrix} 1-5 & 2 \\ 4 & 3-5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation:

$$-4x + 2y = 0 \implies 2y = 4x \implies y = 2x$$

Let $x = k$. Then $y = 2k$.
Eigenvector for $\lambda = 5$ is $X = k \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Example 2: Finding Eigenvalues & Vectors for a 3x3 Matrix

Problem: Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

Step 1: The Characteristic Equation

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 1 & -2 \\ 1 & -\lambda & 0 \\ 0 & 1 & -\lambda \end{vmatrix} = 0$$

Expanding this determinant yields:

$$-\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$$

Solving this cubic equation gives the Eigenvalues: $\lambda = -1, 1, 2$.

Step 2: Eigenvector for $\lambda = 2$ (Let's look closely at this specific calculation)
Substitute $\lambda = 2$ into $(A - \lambda I)X = 0$:

$$\begin{bmatrix} 0 & 1 & -2 \\ 1 & -2 & 0 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

To solve this system, we use row operations to simplify the matrix.
Swap Row 1 and Row 2 ($R_1 \leftrightarrow R_2$):

$$\begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix}$$

Subtract Row 2 from Row 3 ($R_3 \to R_3 - R_2$):

$$\begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

Now, convert this back into algebraic equations:

1. From row 1: $1x - 2y + 0z = 0 \implies x = 2y$
2. From row 2: $0x + 1y - 2z = 0 \implies y = 2z$

Let $z = k$ (a free variable).
If $z = k$, then $y = 2(k) = 2k$.
If $y = 2k$, then $x = 2(2k) = 4k$.

The vector is $X = \begin{bmatrix} 4k \\ 2k \\ k \end{bmatrix}$. Factoring out the k:

Eigenvector for $\lambda = 2$ is $X = k \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$.

(Note: If you are following along with older lecture material that states the answer is $[-1, 1, 1]^T$ for this step, be aware that is a common typo where previous steps were accidentally copied over. The correct mathematical vector for $\lambda = 2$ is $[4, 2, 1]^T$.)

Appendix: The Cross-Multiplication Trick (For 3 Variable Systems)

When finding eigenvectors for 3x3 matrices, you often end up with two equations and three unknowns, like this:

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

Instead of doing tedious row substitutions, you can use the **Cross-Multiplication Method** (a shortcut derived from Cramer's Rule) to instantly find the vector ratios:

$$\frac{x}{(b_1c_2 - b_2c_1)} = \frac{-y}{(a_1c_2 - a_2c_1)} = \frac{z}{(a_1b_2 - a_2b_1)}$$

Quick Example:
If your matrix gives you the equations:
$3x_1 + x_2 + 3x_3 = 0$
$x_1 + 7x_2 + x_3 = 0$

Using the formula:

$$\frac{x_1}{(1)(1) - (7)(3)} = \frac{-x_2}{(3)(1) - (1)(3)} = \frac{x_3}{(3)(7) - (1)(1)}$$

$$\frac{x_1}{-20} = \frac{-x_2}{0} = \frac{x_3}{20}$$

Divide the denominators by 20 to simplify:

$$\frac{x_1}{-1} = \frac{x_2}{0} = \frac{x_3}{1}$$

Your resulting eigenvector is simply the denominators: $X = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.